Artificial Intelligence Chemistry 2 (2024) 100062

Contents lists available at ScienceDirect

Artificial Intelligence Chemistry

journal homepage: www.journals.elsevier.com/artificial-intelligence-chemistry

Machine learning insights into catalyst composition and structural effects on CH_4 selectivity in iron-based fischer tropsch synthesis

Yujun Liu [a,b], Xiaolong Zhang [a,b], Luotong Li [a,b], Xingchen Liu [b,c,*], Tingyu Lei [b,c], Jiawei Bai [b,c], Wenping Guo [d], Yuwei Zhou [d], Xingwu Liu [d,*], Botao Teng [a,*], Xiaodong Wen [b,c,d,*]

[a] Tianjin Key Laboratory of Brine Chemical Engineering and Resource Eco-utilization, College of Chemical Engineering and Materials Science, Tianjin University of Science and Technology, Tianjin 300457, China
[b] State Key Laboratory of Coal Conversion, Institute of Coal Chemistry, Chinese Academy of Sciences, Taiyuan 030001, China
[c] University of Chinese Academy of Sciences, No. 19A Yuquan Road, Beijing 100049, China
[d] National Energy Center for Coal to Clean Fuels, Synfuels China Co., Ltd., Huairou District, Beijing 101400, China

ARTICLE INFO

Keywords:
Artificial Intelligence
Fischer-Tropsch
Catalysis
Methane

ABSTRACT

Fe-based Fischer-Tropsch Synthesis (FTS) enables the selective conversion of syngas into long-chain hydrocarbons, which can be further refined to produce highly demanded liquid fuels and high-value chemical products. However, developing novel heterogeneous catalysts for FTS with desirable performance characteristics is a challenging task, as their performance depends on various factors such as precursor, support material, promoters, pretreatment conditions and the catalyst structures. Thus, it remains difficult to understand the structure-performance relationship of FTS and to optimize the catalyst formulations and operating conditions rationally. By integrating traditional chemistry with machine learning, we herein establish intrinsic correlations among reduction, reaction conditions, phase information and the methane selectivity of Fe-based FTS, using high quality experimental data. The content of the iron phases in the post-reaction phase, particularly χ-Fe_5C_2, significantly influences the methane selectivity of the catalyst. Four types of additives K, Cu, SiO_2, and Ca could effectively suppress the methane selectivity, most likely by promoting or stabilizing the iron carbide phases, indicated by their strong correlation. The machine learned structure-performance relationships offers new insights into the design of Fe-based FTS catalysts, and could guide the further optimization of the preprocessing conditions and various parameter factors to minimize the methane selectivity of FTS.

1. Introduction

Fischer-Tropsch Synthesis (FTS), which was first developed in the early 1900 s, refers to the catalytic conversion of synthesis gas into hydrocarbon compounds, primarily producing alkanes and alkenes, along with the formation of oxygenated compounds [1,2]. FTS offers a viable route for the clean, efficient utilization of coal, marking a significant stride towards sustainable energy solutions. Among the metals that are active as FTS catalysts, iron-based catalysts have high stability, broader temperature and H_2/CO ratio range, and higher selectivity to olefins and C_{5+} hydrocarbons, thus they are deemed to be the best metal for application in industrial scale FTS processes [3].

Iron-based F-T synthesis is a highly complex reaction system, which renders it one of the oldest and perhaps most challenging systems in heterogeneous catalysis [4]. Depending on the pretreatment method, Fe-based FTS catalyst precursors (usually Fe_2O_3 or Fe_3O_4) undergo complicated phase transitions before becoming active in FTS reaction. The optimum catalyst ideally should have high stability, high activity, high C_{5+} selectivity, as well as low CH_4 and CO_2 selectivity. Among them, suppressing CH_4 formation is of paramount importance in FTS, as it is the reverse process of steam reforming, which is used to produce synthesis gas [5].

The performances of FTS are known to be affected by pretreatment and reaction conditions (temperature, pressure, and H_2/CO ratio), the composition of the active phases (χ-Fe_5C_2, ε-Fe_2C, θ-Fe_3C, and Fe_3O_4 etc.) and promoters (Cu, Mn, K, ZnO_2, Al_2O_3, and SiO_2 etc.). Despite the progress in understanding the effect of a few isolated factors on the catalyst performance [6–11], the structure-performance relationship of

* Corresponding authors at: State Key Laboratory of Coal Conversion, Institute of Coal Chemistry, Chinese Academy of Sciences, Taiyuan 030001, China.
* Corresponding authors.
E-mail addresses: liuxingchen@sxicc.ac.cn (X. Liu), liuxingwu@sxicc.ac.cn (X. Liu), tbt@tust.edu.cn (B. Teng), wxd@sxicc.ac.cn (X. Wen).

https://doi.org/10.1016/j.aichem.2024.100062
Received 8 November 2023; Received in revised form 22 March 2024; Accepted 1 April 2024
Available online 2 April 2024

FTS is highly intricate and requires consideration of the effects brought about by multiple factors, which are difficult to optimize simultaneously. Furthermore, studying the structure-performance relationship through experimental means is highly inefficient due to the need for designing, preparing, characterizing, and testing a large number of catalyst formulations and reaction conditions. Consequently, progress in understanding the structure-performance relationship of Fe-based FTS has been relatively slow, and the unified rules for the design of FTS catalysts remains unestablished [12].

To address these challenges, an increasing number of researchers have started applying artificial intelligence (AI) [13] to catalysis research [14–18]. By utilizing AI techniques, catalyst performance can be predicted through machine learning algorithms, thereby accelerating the catalyst design and optimization process [19,20]. Several pioneering studies have already harnessed AI in FTS. *Garona* et al. [21] investigated the production of low-carbon olefins in Fischer-Tropsch synthesis on Co- and Fe-based catalysts using artificial neural networks. By taking input variables such as operating conditions and catalyst composition, and output variables including carbon monoxide conversion rate, light olefin selectivity, and carbon dioxide yield, a three-layer feedforward neural network was adjusted. They found that the neural network provided effective FTS predictions for the most relevant variables, such as temperature and catalyst composition. *Thybaut* et al. analyzed the effect of operating conditions on olefin selectivity and FTS, and suggested that temperature has higher impact on olefin selectivity than pressure and space-time [22]. However, the experimental data on Fe-based FTS reported in the literature usually uses inconsistent protocols of measurements, which can introduce bias or errors that affect the reliability and generalizability of the machine learning model. Moreover, detailed structural characterizations of the catalysts are missing in many literatures, impeding the construction of machine learning models for structure-performances relationships.

In this study, we utilize a set of consistent and high-quality experimental data on Fe-based FTS from a single research team, which has been accumulated over a span of 20 years, to explore the correlations among catalytic conditions, promoter information, and phase information, and methane selectivity using a backpropagation neural network (NN). The identified patterns provide new insights into the structure-selectivity relationship of FTS, and enables the adjusting the pre-processing conditions, reaction conditions, and promoter content to achieve higher performances.

2. Methods

2.1. Construction of databases

The voluminous data amassed over decades of systematic research by a single research team is an invaluable treasure. We extend our profound gratitude to Synfuels China Co. Ltd for their magnanimous support, granting us access to 18 publicly published doctoral dissertations as well as some unpublished foundational data on Fe-based FTS [23–41]. Due to the distinct of Fe with other metals such as Co and Ru in FTS, we herein restrict our analysis to Fe-based catalysts. We have consolidated the catalyst composition, pretreatment conditions, reaction parameters, and catalytic performance into a comprehensive database, primed for machine learning applications. Other journal articles that reported the selectivity of FTS were excluded because of missing key information such as catalyst composition and pretreatment conditions.

The methane selectivity is defined as the weight percentage of CH_4 in the total hydrocarbons of FTS products:

$$S_{CH_4}(\%) = \frac{m_{CH_4}}{m_{Hydrocarbon}} \times 100\%$$

A database was constructed based on the collected data, which consists of 171 records with 40 features and 1 target variable. The various feature parameters in the database are presented in Table 1.

Table 1
Variable summary of the database.

Variable	Variable Type	Possible Values
Reduction Conditions:		
H_2/CO	Input	0.67 / 1.2 / 2
Temperature (K)	Input	250–300
Pressure (MPa)	Input	0.1–1.5
GHSV (h^{-1})	Input	500–7000
Time (h)	Input	13–36
Reaction Conditions:		
H_2/CO	Input	0.67 / 1.2 / 2
Temperature (K)	Input	240–320
Pressure (MPa)	Input	1.5–3
GHSV (h^{-1})	Input	1000–12000
Promoter Contents (%):		
K / Mn / S / SiO_2 / Cu / Ni / Zr /Al / Zn / V / Li / Na / Ca / Rb / Cs / Mg /Sr / Ba / B / Ce / Mo	Input	0–100
Post-Reaction Phase Compositions (%):		
Fe_3O_4 / χ-Fe_5C_2 / ε'-$Fe_{2.2}C$ / Fe^{3+}(spm) / Fe^{2+}(spm) (%)	Input	0–100
CH_4 Selectivity (wt%)	Output	0–100

2.2. Training methodology

The process of machine learning is illustrated in Fig. 1. The database was randomly split into a training set and a testing set using an 8:2 ratio [42], with the random seed set to 42. A neural network framework was then constructed, and the parameters of the neural network were appropriately set. Then, the training set was used to train the model, and the testing set was used for model fitting and evaluating the performance and generalization ability of the model. Finally, a correlation analysis was performed on the features.

The computational work uses Python version 3.11.3, TensorFlow library version 2.12.0, and scikit-learn library version 1.2.1. The training set is imported into the neural network, and the loss function is calculated by comparing the network output with the labels. Commonly used loss functions include cross-entropy loss function and mean squared error (MSE) loss function. Among them, root mean square error (RMSE) can reflect the accuracy of the prediction results [17,43,44]. A smaller RMSE indicates a smaller deviation between the prediction results and the true values, indicating more accurate predictions. RMSE penalizes outliers in the prediction results. Since RMSE calculates the sum of squared errors and then takes the square root, it amplifies the influence of predictions with larger errors on the overall error, thereby more accurately reflecting the overall accuracy of the prediction results. Computing RMSE only requires summing the squared errors and taking the square root, and the unit of RMSE is the same as the unit of the predicted values, allowing for direct comparison of the magnitudes of the predicted values and the true values. Due to these characteristics, the RMSE loss function is used in this study. The formula for calculating RMSE is as follows:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(y_i - y_i')^2}{n}} \tag{1}$$

In the Eq. 1, y represents the actual value, y' represents the predicted value, and n represents the number of samples. The loss function measures the discrepancy between the network output and the labels, and the objective of network training is to minimize the loss function.

Finally, the fitting result is obtained by continuously optimizing and iterating through the gradient descent algorithm [45]. The gradient descent algorithm involves iteratively updating the network along the gradient direction of the loss function with a certain step size. However, the gradient descent algorithm also has some issues. For example, it may converge to a local minimum instead of a global minimum. The convergence speed is slow, especially when approaching the minimum,

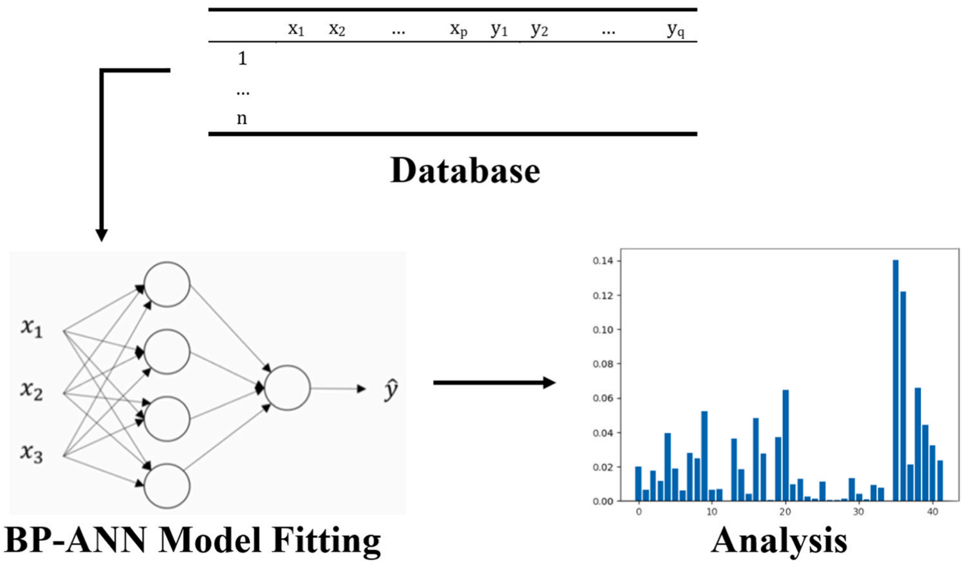

Fig. 1. Machine learning process.

as the gradient decreases and the step size becomes smaller, leading to very slow convergence. Choosing a suitable learning rate is challenging, as the selection of the learning rate has a significant impact on the results, but there is no definitive method to determine the optimal learning rate.

The R-squared (R^2) value, also known as the coefficient of determination, was used to evaluate the degree of model fit [46,47]. The R^2 value is a number between 0 and 1, indicating the correlation between the predicted and actual results of the model and can be used to measure the model's explanatory power for the samples. Specifically, as the R^2 value approaches 1, the model exhibits better explanatory power, indicating a good fit to the sample data. When the R^2 value is close to 0, it suggests that the model fails to explain the variance of the target variable, indicating a poor fit to the sample data. The formula for calculating R^2 is as follows:

$$R^2 = 1 - \frac{sum \quad of \quad squared \quad residuals}{total \quad sum \quad of \quad squares} \tag{2}$$

In the Eq. 2, the "sum of squared residuals" refers to the squared differences between the predicted values and the true values, while the "total sum of squares" refers to the squared differences between the true values and their mean.

Additionally, we introduce another metric, the Mean Absolute Error (MAE) [48], which quantifies the accuracy of machine learning models in predicting. This metric is computed by taking the average of the absolute differences between the predicted values and the true values. The formula for calculating MAE is as follows:

$$E = \frac{\sum_i |Predict \quad Value - Ture \quad Value|}{i} \tag{3}$$

In this Eq. 3, the term "Predict Value" refers to the predicted values, while "True Value" corresponds to the actual values. Through computation in this model, the Mean Absolute Error (MAE) is determined to be 3.69 (rounded to two decimal places).

2.3. Construction of neural networks

A neural network was constructed based on the training set, and the hyperparameters were explored. Setting the number of hidden layers and the number of neurons too large may lead to overfitting, while setting them too small may result in underfitting, both of which are undesirable phenomena [42]. Therefore, it is necessary to determine reasonable values. Through exploration, it was found that setting the number of hidden layers to 3 and the number of neurons per layer to 40 yielded stable Mean Squared Error (MSE) values, as depicted in Fig. 2. This configuration was deemed reasonable and demonstrated accurate performance on the testing set. The iteration count, as shown in Fig. 2, indicated that a stable loss region was achieved at 3000 iterations. However, for this particular sample size, 3000 iterations were already in the range of excessive iterations. Setting a smaller iteration count would result in a larger loss value and poor final fitting performance, which is an aspect currently being improved. The final neural network architecture model, as shown in Fig. 3, consists of 1 input layer with 40 units, 3 hidden layers with 40 neurons each, and 1 output layer with 1 unit. The iteration count is set to 3000. Details of the architecture of the neural network model can be found in the Supporting Information.

The activation function used in the hidden layers of the neural network is "ReLU" (Rectified Linear Unit), and the optimizer is set to "Adam" [49,50]. Compared to functions like sigmoid, the ReLU function significantly reduces computational complexity, resulting in reduced overall computational load. Additionally, the ReLU function sets the output of some neurons to zero, thereby introducing sparsity to the network and reducing interdependent relationships among parameters, mitigating the occurrence of overfitting. The Adam optimizer, known as Adaptive Moment Estimation, is an adaptive optimization algorithm that adjusts the learning rate based on historical gradient information, thereby improving the convergence speed and generalization ability of the model.

To avoid overfitting in constructing the neural network, a dropout function [51,52] was added during fitting. The dropout function randomly drops some neurons during the training process, reducing the dependence between neurons in the neural network and therefore reducing the risk of overfitting. Specifically, the dropout function will drop the output of each neuron according to a certain probability, which is generally between 0.2 and 0.5, and we set it to 0.2. This forces the model to learn more robust features during the training process, thereby improving the model's generalization ability.

3. Results and discussion

3.1. Distribution of methane selectivity in the collected data set

The methane selectivity on Fe-based catalysts is relatively low compared to other metal such as Co and Ru at similar conditions. The frequency distribution of data points with respect to the methane selectivity (Fig. 4) shows three peaks, with one mode centered around

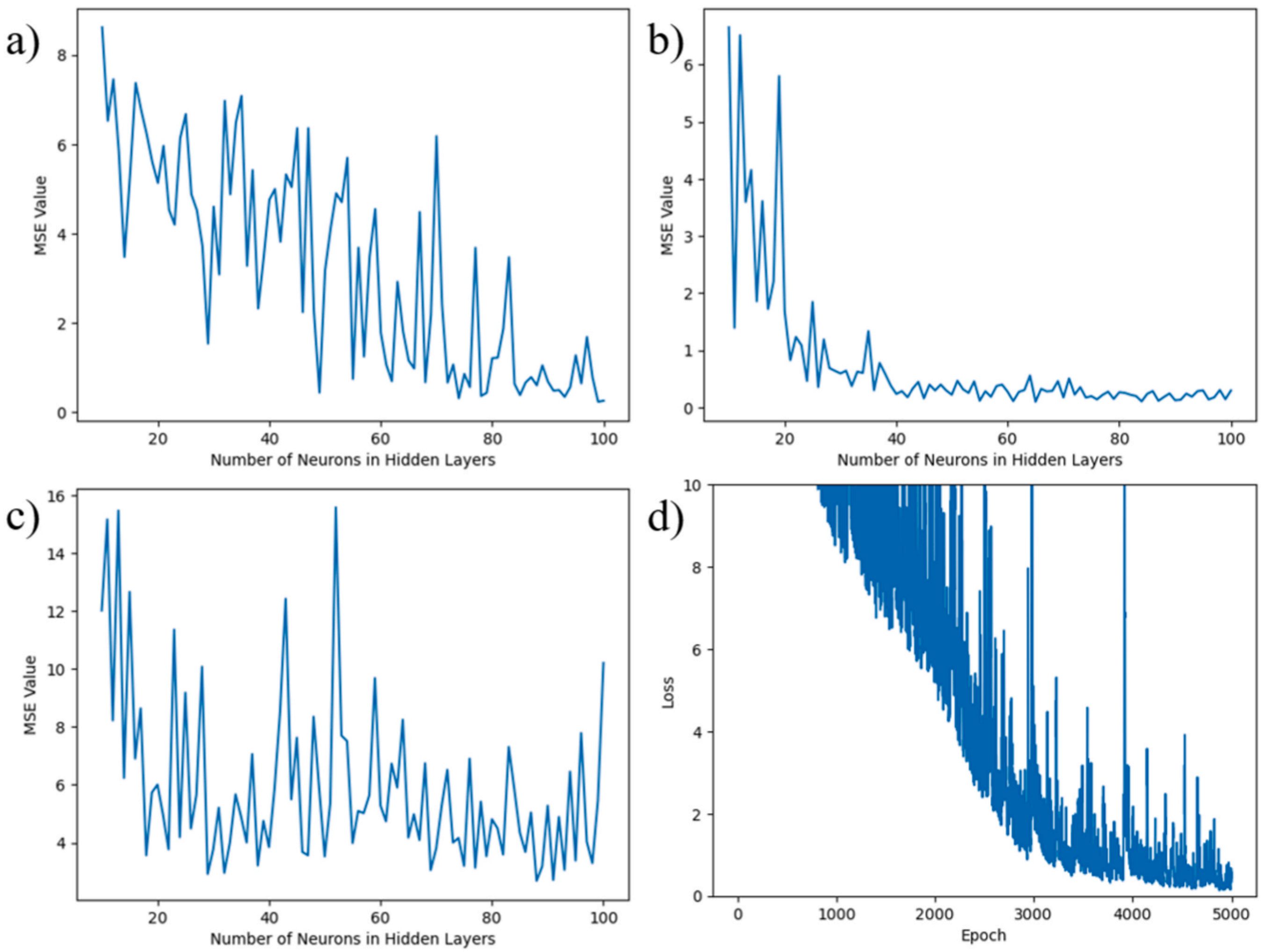

Fig. 2. The minimum MSE (mean squared error) values corresponding to the number of neurons per layer (iteration times are set to 3000). The error is considered reasonable when it approaches stability. a). Two hidden layers, b). Three hidden layers, c). Four hidden layers. d). Relationship between loss value and iteration number.

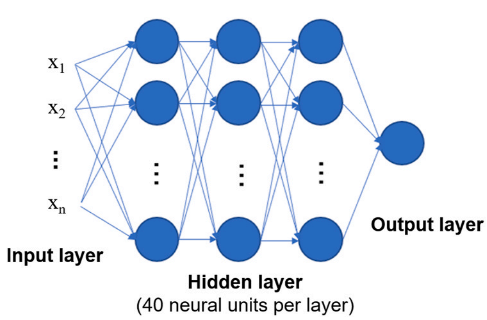

Fig. 3. Neural network architecture.

2–7%, one mode around 15%, and the third one around 22–26%.

In the parameters of the first peak (2–7%), the H_2/CO ratio in the reduction conditions is mostly around 0.67, with a temperature of approximately 270°C (±10°C), pressure mostly below 0.4 MPa, gas hourly space velocity (GHSV) mostly at 1000 h^{-1}, and reduction time mostly around 13 hours. In the reaction conditions, the H_2/CO ratio is mostly 0.67, temperature is around 250°C (±10°C), pressure is mostly 1.5 MPa, and GHSV is mostly 2000 h^{-1}. The catalyst promoters commonly include K (below 6%), Cu (16%±6%), and SiO_2 (5%±3%). The resulting phase information shows a relatively high content of $Fe^{2+/3+}$(spm).

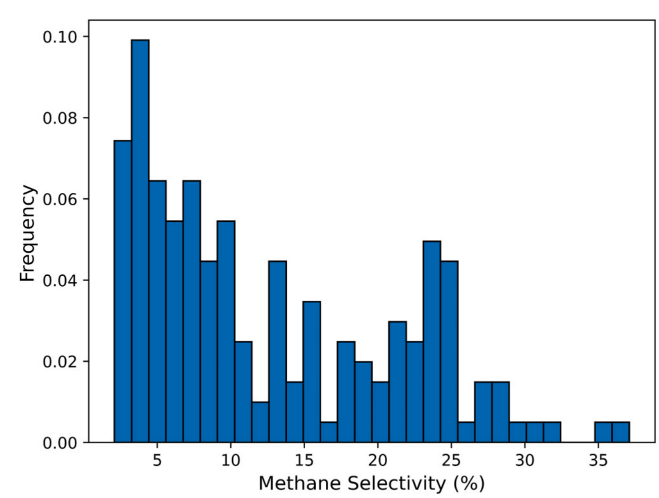

Fig. 4. The frequency distribution histogram of data with respect to methane selectivity in the dataset.

In the parameters of the second peak (approximately 15%), the reduction conditions have a temperature of around 280°C (±10°C), GHSV mostly at 1000 h^{-1}, and reduction time of 20 hours or 24 hours. In the reaction conditions, the temperature is around 270°C (±10°C). The catalyst promoters contain almost no K and Cu.

In the parameters of the third peak (22–26%), the H_2/CO ratio in the reduction conditions is mostly 2, with a temperature around 290°C (±10°C), pressure mostly 0.1 MPa, GHSV mostly at 1000 h^{-1}, and reduction time of 20 hours or 24 hours. In the reaction conditions, the H_2/CO ratio is mostly 2, temperature is mostly 270°C and 280°C, pressure is mostly 1.5 MPa, and GHSV is mostly 2000 h^{-1}. The catalyst does not exhibit significant features. The resulting phase information shows a generally high content of Fe_3O_4 (80%±17%), while the content of $Fe_{2.2}C$ and $Fe^{2+/3+}$(spm) is relatively low (approaching 0).

3.2. Fitting results

The accuracy of the fitted NN model was examined on the testing set in Fig. 5a, showing the differences between the experimental and predicted values of methane selectivity (weight %) in Fe-based FTS. Overall, the predicted values are distributed evenly around the experimental values, with R-squared (R^2) value [17,53] of 0.91 and MAE value of 8.01.

It is commonly known that the bulk or surface structure of the Fe-based catalysts in FTS are subject to drastic change during the pretreating and reaction conditions. Thus, post-reaction phase information may have significant impacts on the CH_4 selectivity. To check this hypothesis, we compared the fitting accuracy by including or excluding the post-reaction phase information (Fig. 5b). Fig. 5a correspond to the fitting results when the post-reaction phase information is included, while Fig. 5b show the results when it is not included. It can be seen from Fig. 5a that the predicted values (blue dots in Fig. 5) are more closely aligned with the experimental values, and are uniformly scattered around the experimental values (red lines in Fig. 5). In contrast, the prediction without the post-reaction phase information is apparently worse (R^2 = 0.73, E=17.53), indicating that the post-reaction phase information has a significant impact on the methane selectivity of Fe-based FTS.

For the fitting results, we also analyzed its residual values [54], as shown in Fig. 6. In Fig. 6a, the residual values between predicted and true values are sorted from small to large, and there is an obvious deviation point where the predicted value has a large difference from the experimental value. This may be due to the fact that some noise points were randomly assigned to the test set during the initial data partitioning, and the feature patterns of these noise points were not learned. This could also be attributed to the insufficient size of the dataset for that particular portion, where the neural network has not yet acquired the underlying patterns. These two factors have resulted in two data points with significant deviations. After removing the data point with large prediction error due to noise interference in Fig. 6b, it can be seen that the residual plot exhibits a clear central symmetric distribution, indicating that the prediction errors of the model conform to a normal distribution. In other words, the model's prediction errors are random, unbiased, and symmetrically distributed on both sides of the true values. This indicates that the model has relatively high prediction accuracy and there is no overall overestimation or underestimation of the predictions. To ensure the consistency and reliability of our findings, we further trained and validated the dataset using the Gradient Boosting Decision Tree (GBDT) method [55]. (Supporting Information).

Overfitting refers to the phenomenon where a machine learning model performs well on the training data but poorly on new, unseen data. Overfitting usually occurs when the model is too complex or the training data is too limited.

To check whether there is overfitting in the trained NN model, K-fold cross-validation method [56,57] was applied. In this method, we divide the dataset into K subsets, and each time one subset is used as the test set and the remaining subsets are used as the training set. This method calculates the performance of the model K times and judges whether the model is overfitting based on these performances. Using K-fold cross-validation allows more accurate evaluation of the performance of the model, because each data point is used to test the model, rather than testing the model with only a portion of the data. We used the 5-fold cross-validation method, which means dividing the data into 5 subsets and judging whether the model is overfitting by calculating the average R^2 value of 5 model runs. The average R^2 value of the cross validation is 0.91, which fluctuates slightly and is close to the R^2 value in the fitting results, indicating that there is no overfitting phenomenon.

3.3. Weight analysis

It has been reported that several factors such as space velocity [58], K promotion [59], and H_2/CO ratio [5] could potentially affect methane selectivity of Fe-based FTS, while CO conversion has little influence [60]. However, a systematic quantification of their contributions is missing. Herein, we analyze the significance of the input parameters on methane selectivity using random forest algorithm. Random forest, as a variant of decision tree algorithm, is a powerful learning algorithm for screening and partitioning big data [61,62]. It is capable of handling high-dimensional datasets, and has good robustness to missing and outlier features. It can capture the interaction between variables,

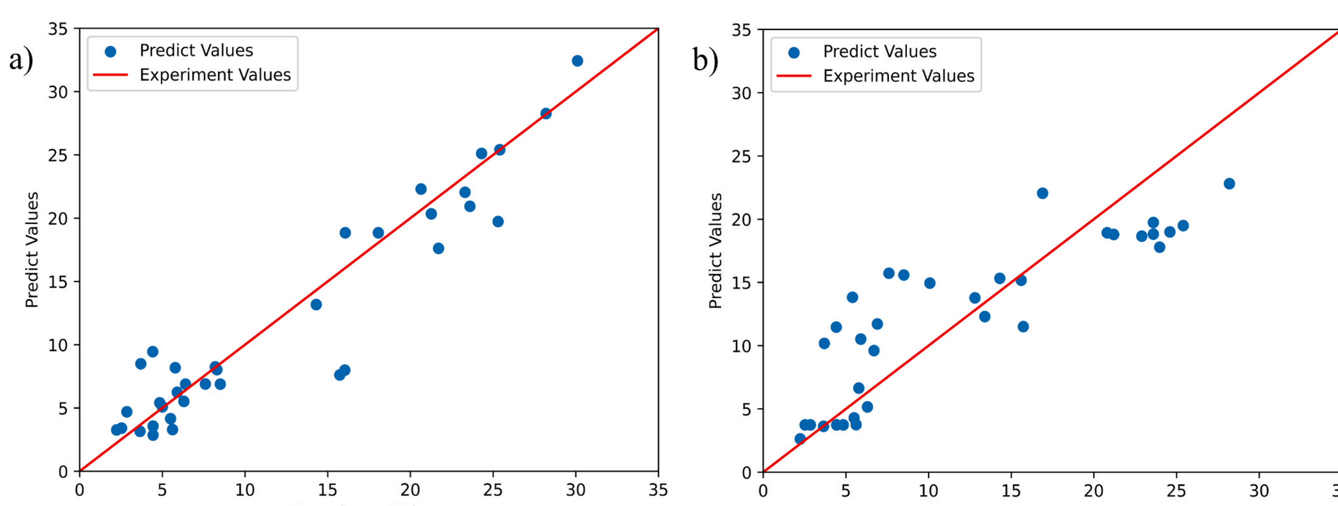

Fig. 5. Comparison of predicted values and experimental values after fitting with neural network. a). Including post reaction phase information. b). Excluding post reaction phase information.).

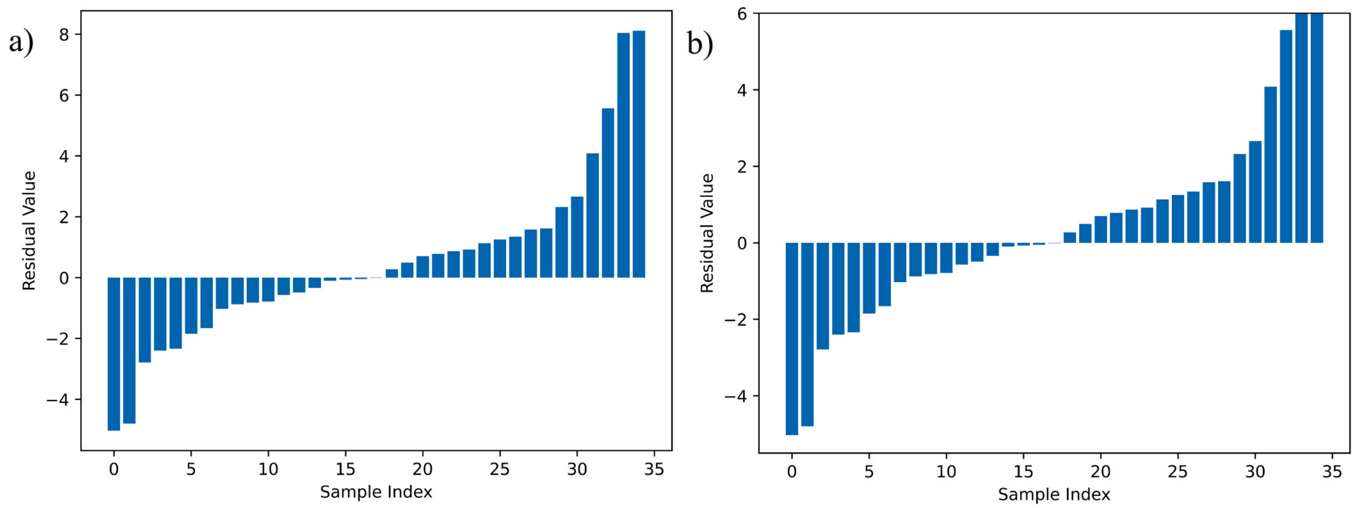

Fig. 6. Residual plot, displaying the difference between predicted and experimental values sorted from small to large. a) the residual plot after fitting, and b) the residual plot obtained after removing the data point with large prediction error due to noise interference.

making it suitable for handling non-linear relationships. Random forest can handle high-dimensional data without feature selection, as it can automatically select the most important features. It can be used for classification and regression problems, and the model's bias and variance can be balanced by adjusting its parameters to improve its generalization performance.

We divided the data into 9 classes with an average interval of 3.9 between 2.02 and 371 [63,64]. The drawback of this size-based division is that the frequency distribution of each class is uneven, which can lead to inaccurate predictions by the random forest. To address this, we expanded the initial data by replicating the classes with lower frequencies to achieve nearly equal frequency distribution in each class. The data was then randomly split into training and testing sets in an 8:2 ratio, a decision based on literature [65] that aligns with best practices for datasets of our type and size. In Fig. 7, the horizontal axis represents the sequence number of the sample features, corresponding to the input feature parameters shown in Table 2, and the vertical axis represents the weight proportion. It can be inferred from the figure that the significance of the post-reaction phase is far ahead of other feature parameters. In particular, the Fe_3O_4 and χ-Fe_5C_2 content in post-reaction phase have the highest weight proportions, indicating that they may play a decisive role in controlling methane selectivity. It is well known that methane molecules are formed on the iron carbide phases through hydrogenation of the adsorbed CH_x species, while Fe_3O_4 is inactive in methane formation. Considering that the sum of the iron contents remains constant during the reaction, the content of the iron phases in the post-reaction phase, particularly χ-Fe_5C_2, influences the methane selectivity of the catalyst most significantly among all the input features. Therefore, tuning the methane selectivity of the Fe-based FTS should primarily focus on the phase engineering of the iron catalysts to promote the content of χ-Fe_5C_2. Other feature parameters with high weight proportions include the Fe^{3+}(spm) in the post-reaction phase, the content of some additives, the reduction temperature, and the reaction temperature.

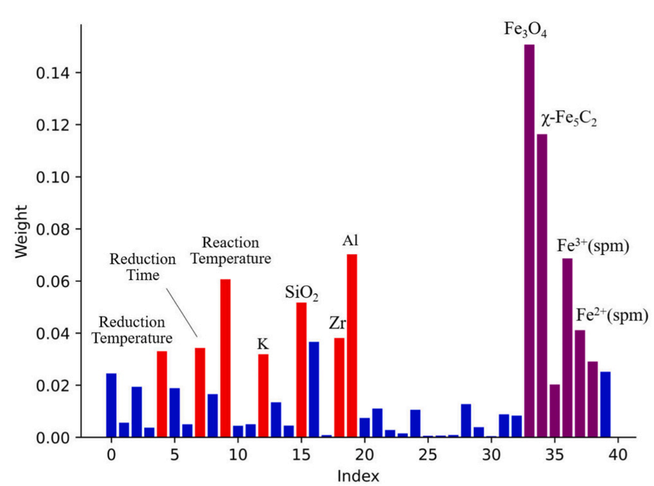

Fig. 7. Weight proportions of the impact of the input parameters on the results in the random forest study. The red bars represent features with relatively high weight proportions, and the purple bars represent post-reaction phase components features with relatively high weight proportions; other features are colored in blue.

3.4. Correlation analysis

Although there are 39 input feature parameters in this neural network, there are many related features among them. For example, the reduction and reaction conditions are expected to have a direct relationship with the post-reaction phase information. In order to investigate the correlation between these input feature parameters, Pearson correlation analysis was performed [66]. The Pearson correlation coefficient is a statistical method for measuring the linear correlation between two variables. It can be used to evaluate the correlation between two variables, that is, how one variable changes when the other variable changes [67]. The value of the Pearson correlation coefficient ranges from −1–1. A correlation coefficient of 1 indicates that the two variables are perfectly positively correlated; while a value of −1 indicates perfectly negatively correlation. A value of 0 indicates that the two variables are not linearly correlated. Pearson correlation coefficients are calculated using the following formula:

$$r(X,Y) = \frac{Cov(X,Y)}{SD(X) \bullet SD(Y)} \quad (4)$$

In the Eq. 4, Cov(X,Y) represents the covariance between variable X and variable Y, and SD(X) and SD(Y) represent the standard deviation of variable X and variable Y, respectively.

The heatmap of the Pearson correlation coefficient of the database is shown in Fig. 8. It can be clearly seen that the content of Fe_xC and Fe ions in the post-reaction phase information has a strong negative correlation with the selectivity of CH_4, and has a strong negative correlation with Fe_3O_4 in the post-reaction phase information. This suggests that a higher concentration of Fe_xC and Fe ions in the post-reaction phase significantly

Table 2
Input feature parameters corresponding to the serial numbers (Due to the preprocessing condition where only H_2 is used for the first half of the time and only CO is used for the last half of the time, the H_2/CO ratio cannot be represented. Therefore, four variables are used to describe this feature. When only H_2 is used, the value is set to 999, and the same applies when only CO is used. The 40th feature represents the selectivity for CH_4.).

Reduction Conditions									Reaction Conditions			
Index	0	1	2	3	4	5	6	7	8	9	10	11
Corresponding Information	First Half H_2	CO	Last Half H_2	CO	Temperature (°C)	Pressure (MPa)	GHSV (h^{-1})	Time (h)	H_2/CO	Temperature (°C)	Pressure (MPa)	GHSV (h^{-1})
Promoter Contents(%)												
Index	12	13	14	15	16	17	18	19	20	21	22	
Corresponding Information	K	Mn	S	SiO_2	Cu	Ni	Zr	Al	Zn	V	Li	
Index	23	24	25	26	27	28	29	30	31	32		
Corresponding Information	Na	Ca	Rb	Cs	Mg	Sr	Ba	B	Ce	Mo		
Post-Reaction Phase Compositions(%)												
Index	33	34	35	36	37	38	39					
Corresponding Information	Fe_3O_4	χ-Fe_5C_2	ε'-$Fe_{2.2}C$	Fe^{3+}(spm)	Fe^{2+}(spm)	Fe_xC	$Fe^{2+/3+}$(spm)					

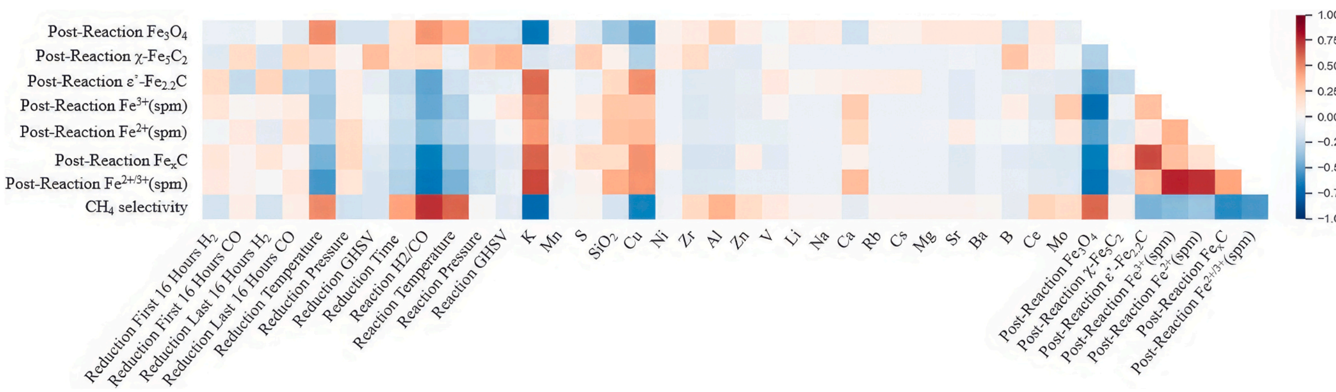

Fig. 8. Heatmap of Pearson correlation coefficients.

enhances the selectivity towards CH_4. This enhancement is primarily due to the superior C-C coupling capability of Fe_xC species, especially χ-Fe_5C_2, which amplifies the chain growth probability during FTS [68, 69].

The reduction temperature, reaction temperature and the H_2/CO ratio in the reaction conditions all have a clearly positive correlation with CH_4 selectivity and the Fe_3O_4 content in post-reaction catalyst phase, while having a negative correlation with Fe_xC and Fe ions in the post-reaction catalyst. This suggests that a higher reduction temperature and H_2/CO ratio in the reaction conditions can inhibit the formation of Fe_xC and Fe ions in the working conditions and promote the formation of CH_4.

Traditional Fischer-Tropsch synthesis (FTS) research widely acknowledges the significant role of alkali metal promoters (such as K, Na, Li, Rb, and Cs) in suppressing methane selectivity [8,70]. To some extent, alkaline earth metal promoters (such as Mg and Ca) also influence the suppression of methane selectivity [71,72]. The composition of catalyst phases affects the selectivity for methane and other products, although the specific patterns remain unclear. Our findings reveal that four types of additives K, Cu, SiO_2, and Ca effectively suppress methane selectivity. Additionally, we found a clear positive correlation between the K promoter and the post-reaction Fe_xC content. This suggests that K acts not only as an electronic promoter to suppress methane selectivity, as traditionally believed, but also as a structural promoter stabilizing carbide phases [73,74]. The observations regarding K and Ca's roles in suppressing methane selectivity align with traditional perspectives. Furthermore, our AI-based analysis indicates that Cu and SiO_2, typically deemed to minimally impact FTS product selectivity, also play similar roles as K and Ca in suppressing methane selectivity. This insight might explain why these two promoters are frequently included in industrial catalyst formulations.

Using random forest and Pearson correlation coefficient analysis, we discerned that the information pertaining to the post-reaction phase significantly influences CH_4 selectivity. Specifically, the concentration of χ-Fe_5C_2 and Fe ions within the post-reaction phase emerges as the paramount factors impacting CH_4 selectivity. Therefore, achieving low methane selectivity is possible by enhancing the proportion of the carbide phases in the post-reaction catalyst. Furthermore, the reduction conditions, reaction conditions, and additive content indirectly influence CH_4 selectivity, likely through modifications to the phase composition of iron catalysts in Fischer-Tropsch Synthesis (FTS). On a quantitative level, the optimal conditions for methane selectivity in iron-based FTS can theoretically be predicted by employing optimization algorithms, such as genetic algorithms, in tandem with neural-network models. Our practical findings indicate that allowing all experimental parameters to vary makes it challenging for the genetic algorithm to converge, given the vast parameter space. Nonetheless, by holding certain less critical parameters constant, we can achieve optimal conditions for methane selectivity through a successful optimization process (Fig. 9). Specifically, the optimization converges on the following parameters: under reduction conditions, a H_2/CO ratio of 0.67, a reaction temperature of 300°C, a pressure of 0.1 MPa, and a flow rate of 1000 h^{-1}; under reaction conditions, the H_2/CO ratio remains at 0.67, but the temperature is increased to 320°C, pressure to 1.5 MPa, and the flow rate is doubled to 2000 h^{-1}. Additionally, the catalyst composition is set at 100Fe/5Cu/6 K/16SiO_2 (wt/wt). With these optimized conditions, the post-reaction phase composition includes 57.46% Fe_3O_4, 2.87% χ-Fe_5C_2, 34.26% ε'-$Fe_{2.2}C$, and 16.47% $Fe^{2+/3+}$(spm), leading to a methane selectivity of 2.98%. This detailed information could facilitate the design of Fe-based FTS catalysts that achieve low methane selectivity.

4. Conclusions

In this work, we have used neural networks to fit and analyze the experimental parameters and the methane selectivity of Fischer-Tropsch Synthesis. Our results reveal clear relationships among reduction conditions, reaction conditions, post-reaction phase information, and CH_4 selectivity. Most notably, we have shown that the post-reaction phase of the iron catalyst, particularly χ-Fe_5C_2, is the most significant factor that determines the methane selectivity of Fe-based FTS. Thus, tuning the oxide/carbide ratio of the Fe phases in the reaction condition is the key to suppress methane formation. Four types of promoters K, Cu, Ca, and SiO_2 could promote or stabilize the χ-Fe_5C_2 phase, thereby reduce methane selectivity. Moreover, methane selectivity could be suppressed by reducing the reduction and reaction temperature and the H_2/CO ratio of Fe-based FTS. Such insights would shed important light on the design of Fe-based FTS with low methane selectivity. However, machine learning also faces numerous challenges when predicting complex systems like Fischer-Tropsch synthesis, making it challenging to achieve highly accurate predictions. Possible reasons for this may include significant noise interference and insufficient of data in the database due to the high cost of characterizing the composition and structure of the catalysts.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xiaodong Wen, Xingchen Liu, Xingwu Liu, Botao Teng reports financial support was provided by National Natural Science Foundation of China. Xiaodong Wen reports financial support was provided by Chinese Academy of Sciences. Xiaodong Wen reports financial support was provided by Ministry of Science and Technology of the People's Republic of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors are grateful for the financial support from National Science Fund for Distinguished Young Scholars of China (Grant No. 22225206), National Natural Science Foundation of China (Nos. 22372187, 22172183, 22372120, 22302220, 21972160), CAS Project for Young Scientists in Basic Research (YSBR-005), Key Research Program of Frontier Sciences CAS (ZDBS-LY-7007), Major Research plan of the National Natural Science Foundation of China (92045303), the Informatization Plan of Chinese Academy of Sciences (Grant No. CAS-WX2021SF0110), Ordos key research and development program (NO. YF20232317), Youth Innovation Promotion Association CAS (2020179), and Shanxi Province Science Foundation for Youth (No. 202203021222403). This work was also financially supported by the autonomous research project of SKLCC (Grant No.: 2023BWZ005).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.aichem.2024.100062.

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